

Chin Zhe Ning

Singapore

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EDUCATION

National University of Singapore

College of Humanities and Sciences · (Aug 2022 - May 2026)

- Bachelor of Science: (Hons) Double Major in Mathematics (1st) and Statistics (2nd), with Minor in Computer Science
- Highest Distinction
- Awards: AY25/26 Semester 1 Dean's List
- Relevant Coursework: Optimization for Large-scale Data-driven Inference, Data Modelling and Computation, Applied Time Series Analysis, Nonlinear Programming, Statistical Computing, Mathematical Statistics, Software Engineering

PROJECTS

Monte Carlo Methods and their Applications in Finance

Capstone project supervised by Prof. Ren Weiqing. · [REPO](#)

- Applied variance reduction techniques for Monte Carlo in rare-event settings such as deep-OTM option pricing and credit risk default probability estimation under multi-factor normal copula models.
- Analyzed estimator efficiency and variance behaviour under extremely low-probability events, with control variates, importance sampling, and control functionals.

Simulation-based Inference in Adaptive Network Epidemic Models

Project for the course Simulation (ST3247) taken under A. Prof. Alexander Thiery. · [REPO](#)

- Applied Approximate Bayesian Computation to fit adaptive epidemic network models with rewiring dynamics to opaque timeseries data.
- Evaluated automatic summary statistic design and calibration quality across inference settings.
- Normalizing flows (SDE model) was used to learn a posterior directly for amortized inference.
- Rewrote the model simulator in C++ for $75\times$ speed-up over the base Python implementation.

WORK EXPERIENCE

Resident Assistant

Ridge View Residential College · (Aug 2024 - May 2026)

- Maintained a safe and inclusive residential community through student engagement and support.
- Acted with residential life staff to respond to crisis situations and ensure timely resolution and follow-up.

Volunteer Tutor

Project Lightbulb · (May 2024 - Dec 2025)

- Provided Mathematics and Science tutoring to primary and secondary school students in Singapore to support academic development and learning outcomes.
- Mentored students from underserved backgrounds referred through the Ministry of Social and Family Development and community partners.

Teaching Assistant

School of Computing · (Aug 2023 - Dec 2023)

- Conducted weekly tutorials for the course CS1010S: Programming Methodology I, an introduction to programming in Python.
- Break down hard concepts (e.g. recursion) into simple, intuitive explanations and examples.
- Student feedback: 4.20/5.00

SKILLS

- Programming & Tools: Python, SQL, R, LaTeX, Typst
- MLOps: PySpark, Hadoop, Snowflake
- Scientific Computing: NumPy, SciPy, Pandas, matplotlib, PyTorch and Jax.

Coursera

- SQL for Data Science (UC Davis)

- Spark, Hadoop, and Snowflake for Data Engineering (Duke)
- Bayesian Statistics: From Concept to Data Analysis (UC Santa Cruz)